

AI-Driven 3D Organism Growth Simulation

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Background: Modeling Biological Growth with AI

- • Biological systems grow in branching, spiral, and gradient-based patterns.
- • AI-driven simulations replicate these patterns for synthetic biology applications.
- • Goal: Develop a self-organizing 3D growth model using Neural Cellular Automata (NCA).

Methods: AI-Driven Growth Simulation

- • Growth modeled using Neural Cellular Automata (NCA) and reinforcement learning.
- • Integrated three growth types:
 - - Branching (L-systems, Diffusion-Limited Aggregation)
 - - Spiral (Fibonacci-based patterns)
 - - Gradient-based (morphogen diffusion, biomechanical feedback)
- • Implemented in Python with PyTorch and Trimesh for 3D visualization.

Results: AI-Guided Growth Patterns

- • Successfully generated branching networks, spiral formations, and gradient-adaptive structures.
- • Growth self-organized without predefined structures, responding to simulated biomechanical forces.
- • Demonstrated potential for bioengineering applications such as synthetic tissue design.

Conclusion & Future Directions

- • AI-driven growth models replicate natural biological development.
- • Future work:
 - - Incorporate real-world material properties for improved biomechanical realism.
 - - Map AI-generated growth patterns to DNA sequences for synthetic biology applications.
 - - Explore applications in regenerative medicine, bio-printed tissues, and self-growing architecture.